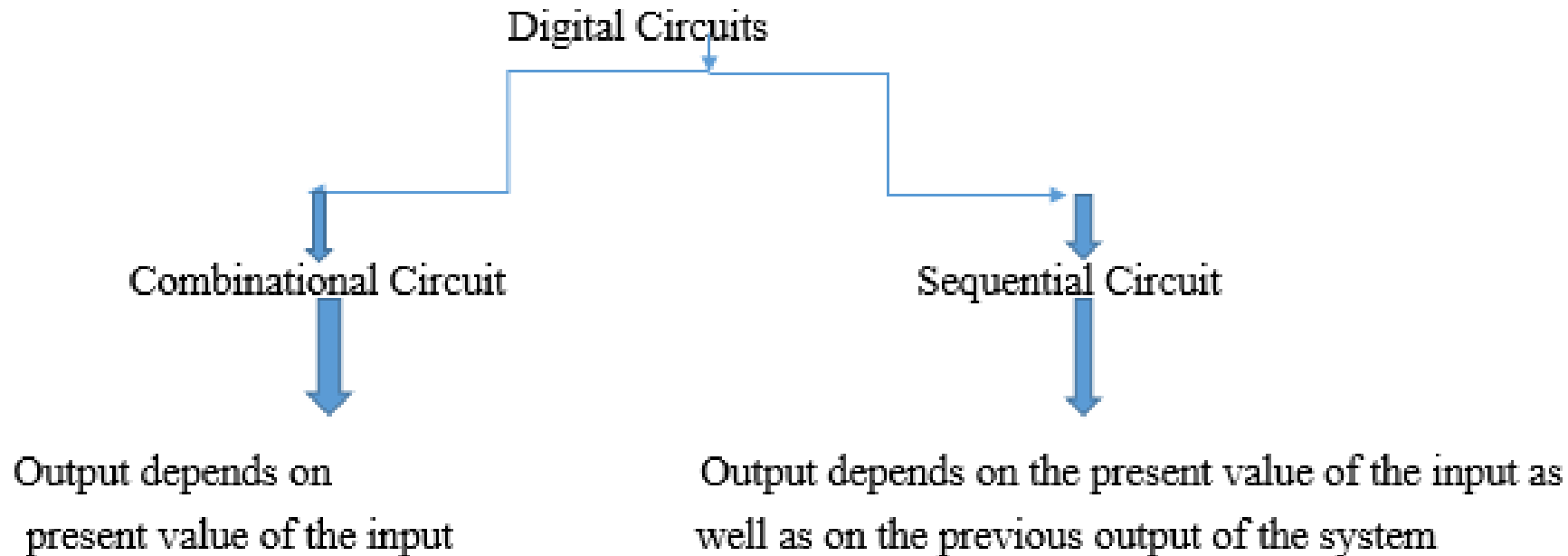


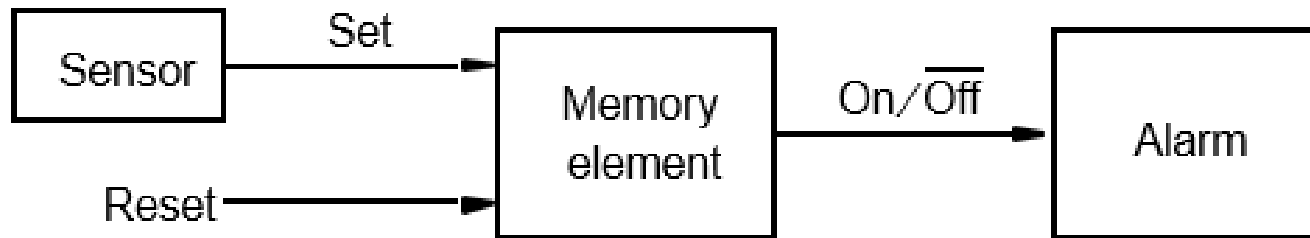
23ECE205
Digital Electronics and Systems
III Semester
B. Tech. (CSE)
Amrita School of Engineering, Bengaluru
Lecture

Combinational Circuit and Sequential Circuit



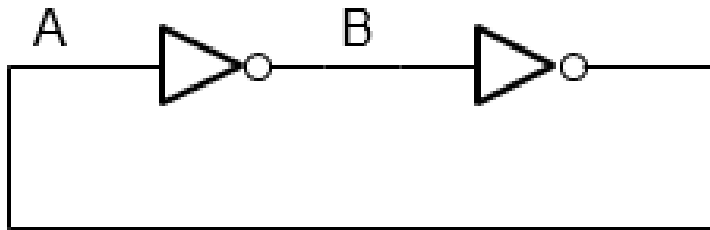
Sequential circuits include storage elements that store the values of the previous state of the system. The contents of the storage elements are said to represent the *state* of the circuit.

Control of an Alarm System



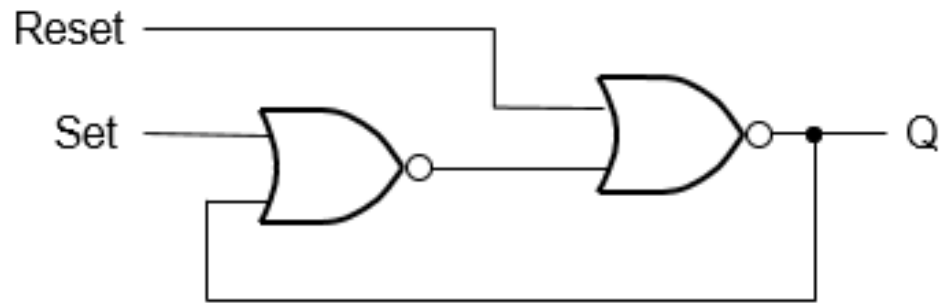
The alarm mechanism responds to the control input *On/Off'*. It is turned on when $On/Off' = 1$, and it is off when $On/Off' = 0$. The desired operation is that the alarm turns on when the sensor generates a positive voltage signal, *Set*, in response to some undesirable event. Once the alarm is triggered, it must remain active even if the sensor output goes back to zero. The alarm is turned off manually by means of a *Reset* input. The circuit requires a memory element to remember that the alarm has to be active until the *Reset* signal arrives.

A Simple Memory Element



If we assume that $A = 0$, then $B = 1$. The circuit will maintain these values indefinitely. If we assume that $A = 1$, then $B = 0$, and the circuit will remain in this second state indefinitely. Thus, the circuit has two possible states. This circuit is not useful, because it lacks some practical means for changing its state.

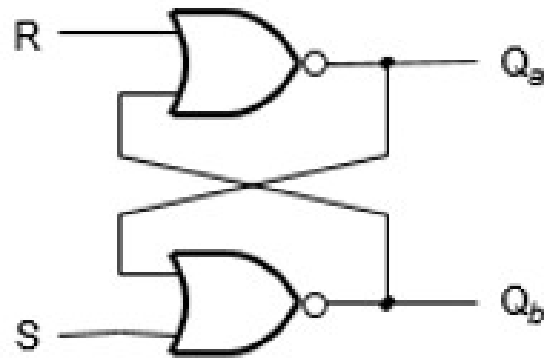
A Memory Element with NOR gates



Set (S)	Reset (R)	Q	Remarks
0	0	Q	Previous State
0	1	0	Reset State
1	0	1	Set State
1	1	0	But this is invalid state which will be discussed soon

A Latch built with NOR gates

Required: Q_a and Q_b are complement of each other



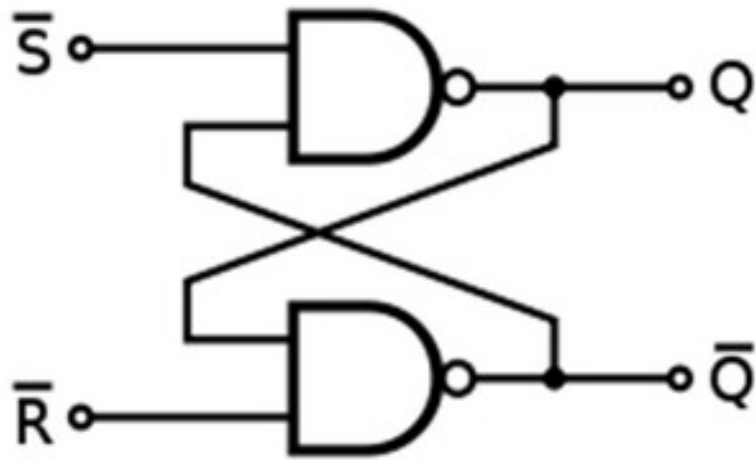
(a) Circuit

S	R	Q_a	Q_b	
0	0	0/1	1/0	(no change)
0	1	0	1	➡ Reset State
1	0	1	0	➡ Set State
1	1	0	0	➡ Invalid

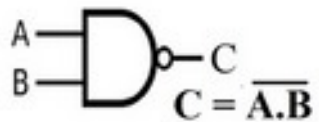
(b) Truth table

A Latch built with NAND gates

The next state Q can be written as either $Q(n+1)$ or $Q(t+1)$ or $Q+$
 The next state Q' can be written as either $Q'(n+1)$ or $Q'(t+1)$ or $Q'+$ } different authors follow different notations



NAND GATE

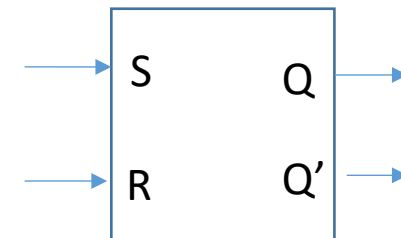


Truth Table

INPUT		OUTPUT
A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0

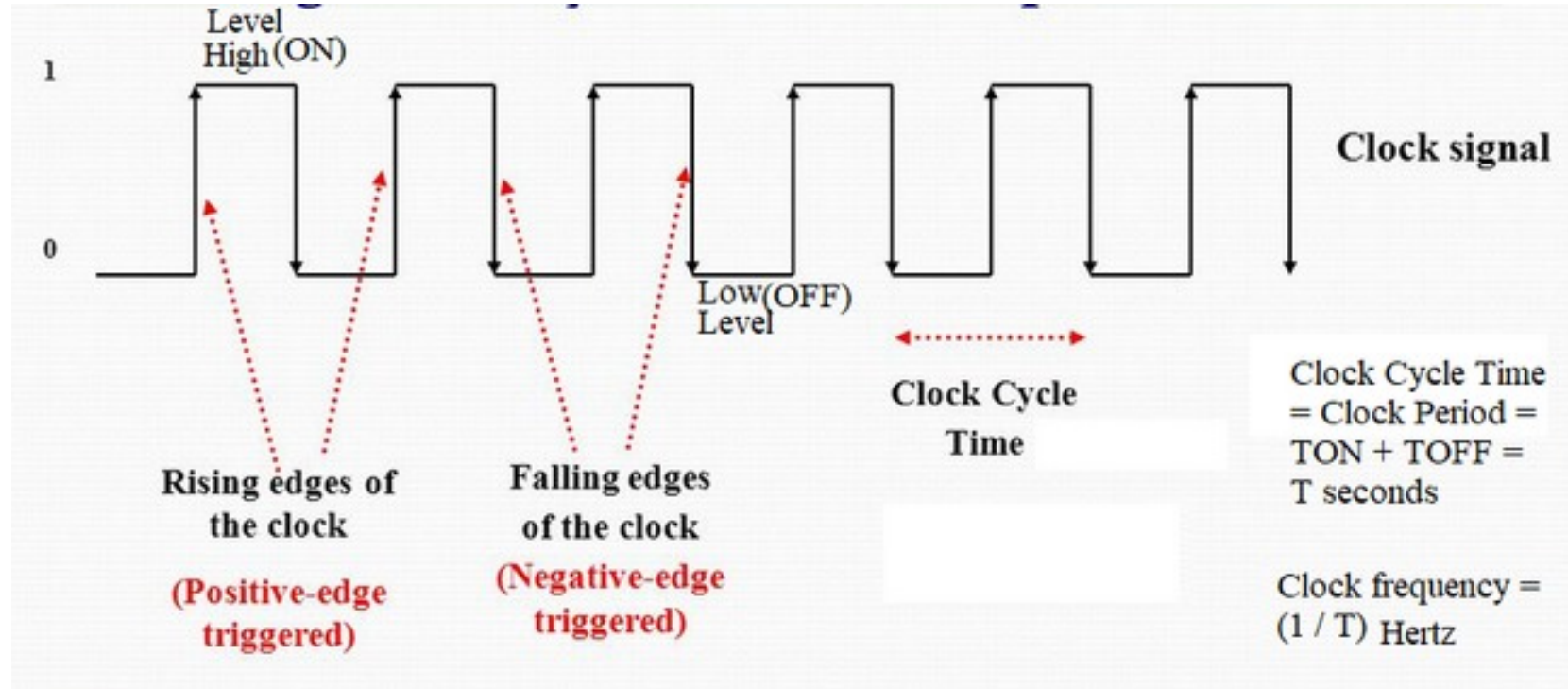
S	R	S'	R'	Q	Q'	Remarks
0	0	1	1	Q	Q'	Same Previous value
0	1	1	0	0	1	Reset State
1	0	0	1	1	0	Set State
1	1	0	0	1	1	Not Allowed

Symbol of SR Latch

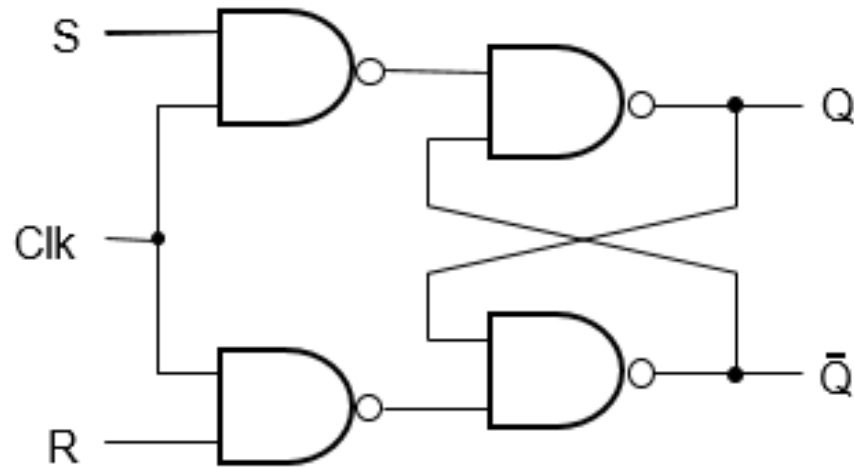


Clock

- A clock is that which continuously alternates between 0 and 1. i.e. it is continuous train of pulses
- The time it takes the clock to change from 1 to 0 and back to 1 (or vice versa) is called the clock period or clock cycle time.
- i.e. Cycle time of a clock is the sum of its high duration and its low duration. The length of the time the clock is high before changing states is its high duration and the length of the time the clock is low before changing states is its low duration
- The clock frequency is the inverse of the clock period. The unit of measurement for frequency is the Hertz
- Clocks are often used to synchronize circuits

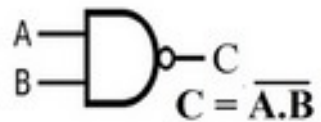


Gated SR Latch with NAND gates



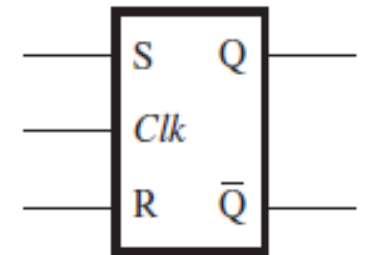
Clk	S	R	Q(t+1)	Q'(t+1)	Remarks
0	x	x	Q(t)	Q'(t)	No change. Same as that of the previous state
1	0	0	Q(t)	Q'(t)	No change. Same as that of the previous state
1	0	1	0	1	Reset state
1	1	0	1	0	Set state
1	1	1	1	1	Invalid

NAND GATE



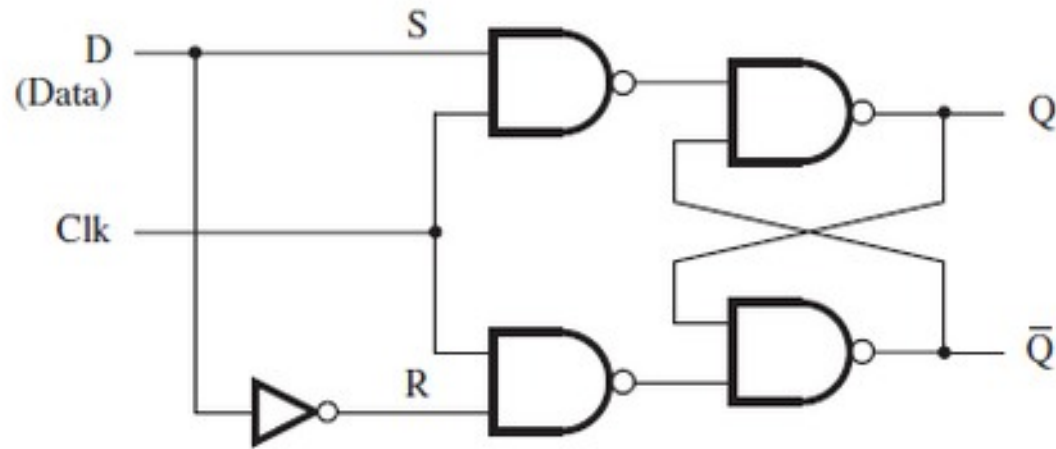
Truth Table

INPUT		OUTPUT
A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0



(d) Graphical symbol

Gated D (Data) Latch

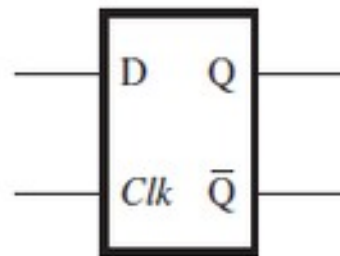


(a) Circuit

Clk	D	Q	Q'	Remarks
0	x	Q	Q'	No change. Same as that of the previous state
1	0	0	1	Q will be same as that of D input
1	1	1	0	Q will be same as that of D input

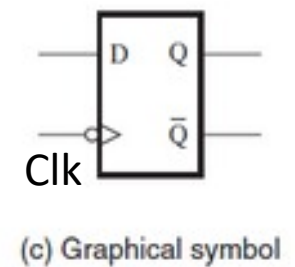
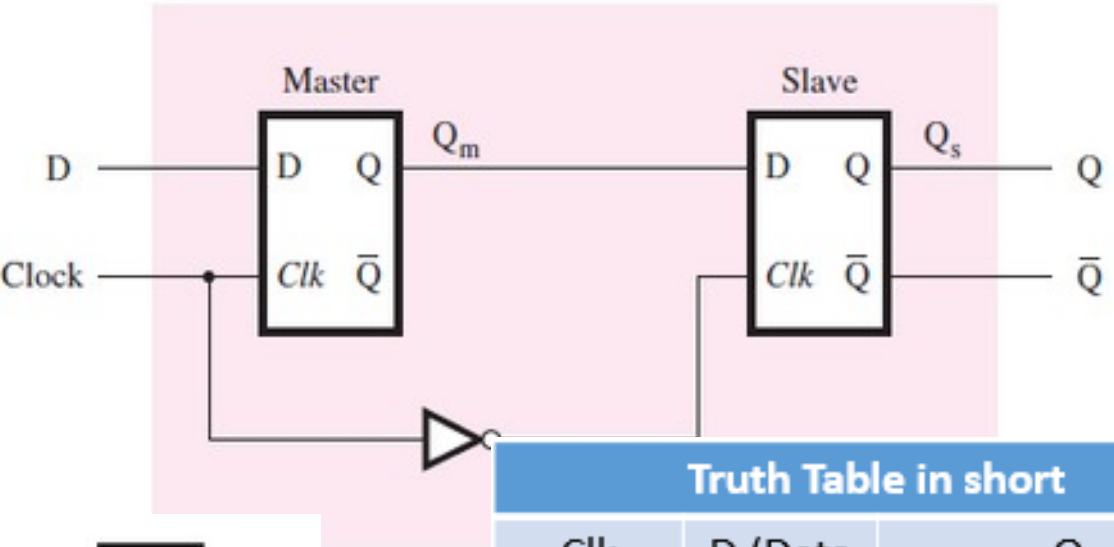
Clk	D	Q(t + 1)
0	x	Q(t)
1	0	0
1	1	1

(b) Truth table



(c) Graphical symbol

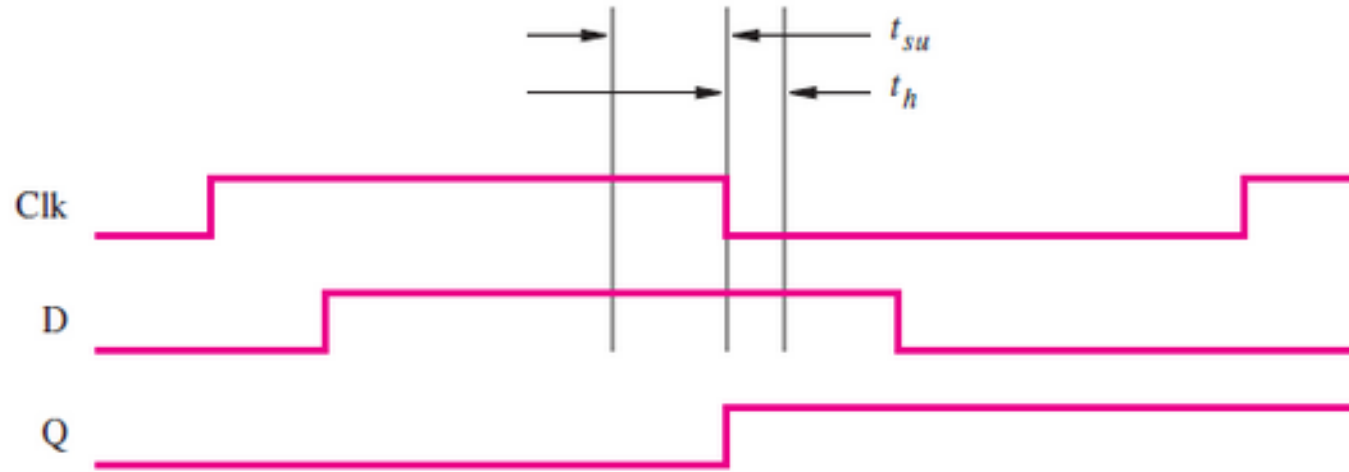
Master Slave D Flip Flop (Negative Edge Triggered D Flip Flop)



Truth Table in short		
Clk	D (Data Input)	Q
	D	Q = D
Any other value on clock	x	Q = same as that of Previous value

Truth Table in a descriptive manner					
Clk		D	Qm	Qs = Q	Remarks
1	-ve edge on Clk	1	1	Previous output = Q	At negative edge, whatever is value of D data input the same value will appear on Q
0		x	1	1	
	-ve edge on Clk	0	0	Previous output = Q	At negative edge, whatever is value of D data input the same value will appear on Q
		x	0	0	
	x	x	Qm	Q	At any other value other than negative edge trigger on the clock, the same previous value of Q will continue on Q

Set Up Time and Hold Time



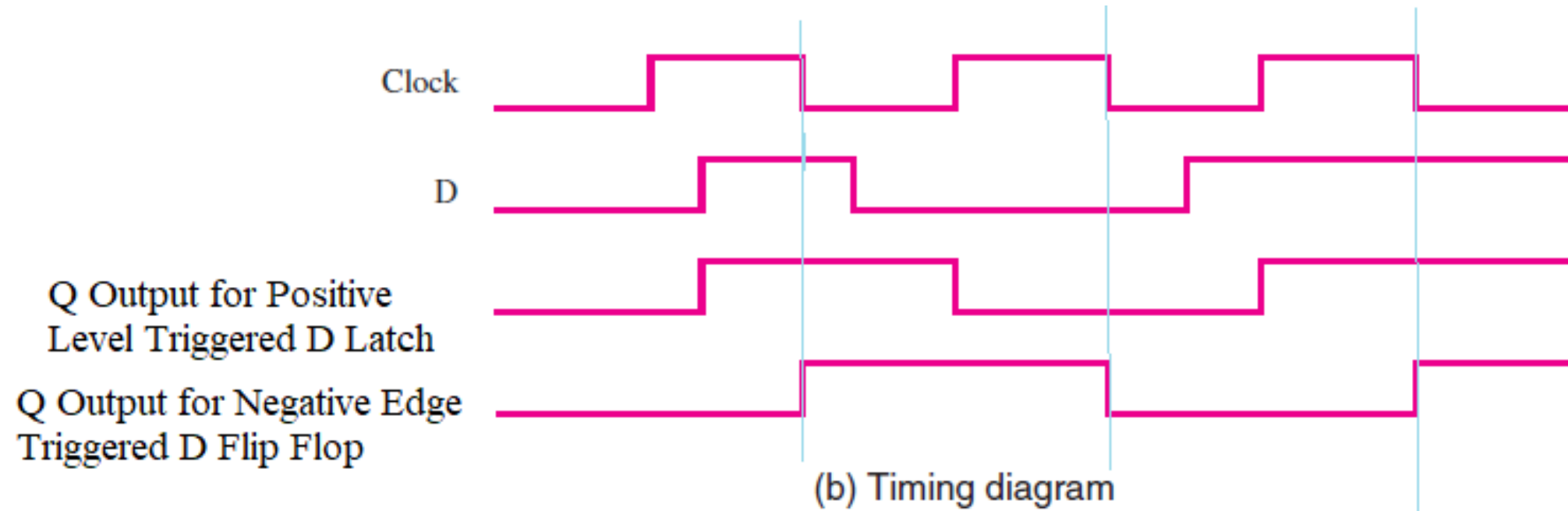
Set Up Time (t_{su}): The minimum time that the D signal must be stable prior to the negative edge (or positive edge based on what type of flipflop) of the Clk signal is called the *setup time*, t_{su} , of the latch.

Hold Time (t_h): The minimum time that the D signal must remain stable after the negative edge (or positive edge based on what type of flipflop) of the Clk signal is called the *hold time*, t_h , of the latch.

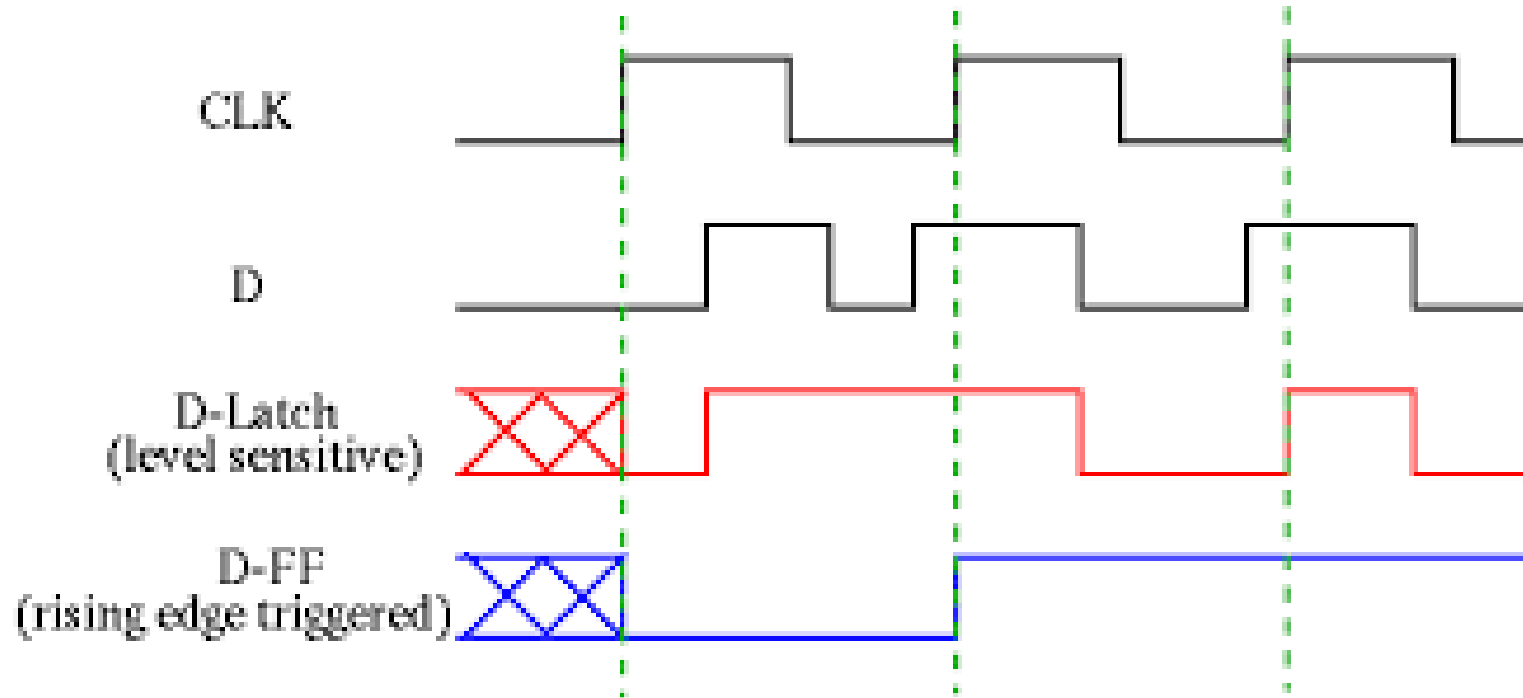
Typical values for $t_{su} = 3$ ns and $t_h = 2$ ns.

If Set up time and Hold time are not adhered then the circuit will enter in Metastable state.

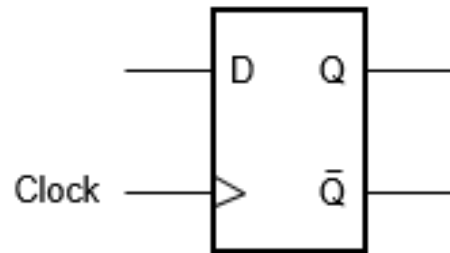
Timing Diagram Showing Difference between D Latch and D Flip Flop Working



Analyze this Waveform: Exercise



A Positive-Edge-Triggered D flip-flop.

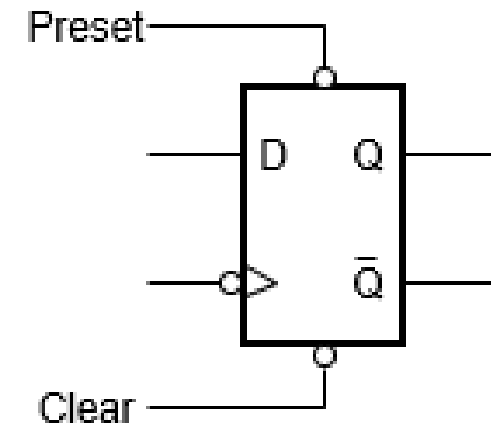


(b) Graphical symbol

Master-Slave D flip-flop with Clear and Preset

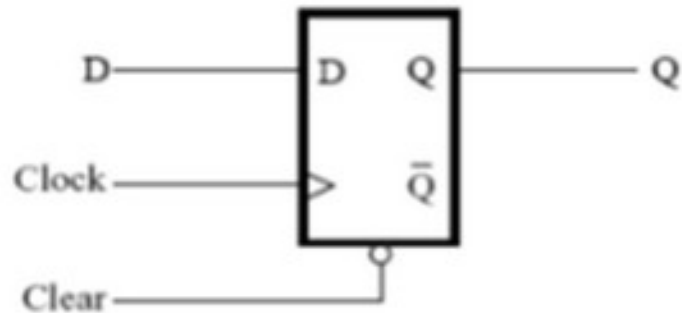
Preset	Clear	D Input	Q Output	Remarks
0	0	Don't Care	Don't Care	Invalid
0	1	Don't Care	1	Preset Condition
1	0	Don't Care	0	Clear or Reset Condition
1	1	Whatever Valid value on D Input	Q = D Valid Output on Q	Regular or Normal Functionality of D Flip flop begins

We have assumed Low Level Active Preset and Clear here.
* Practically, these may be Low **level** or High level active



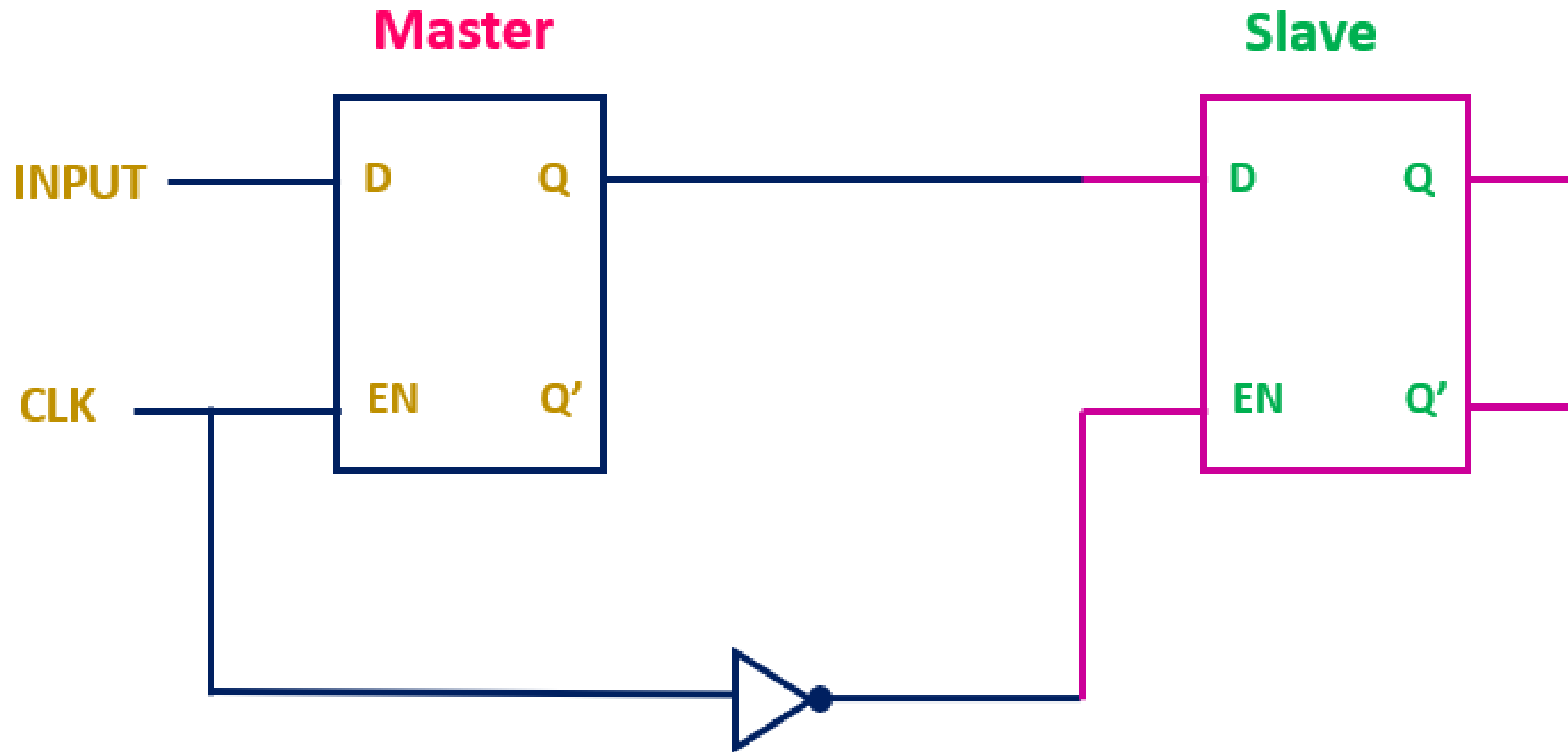
(b) Graphical symbol

Asynchronous Clear

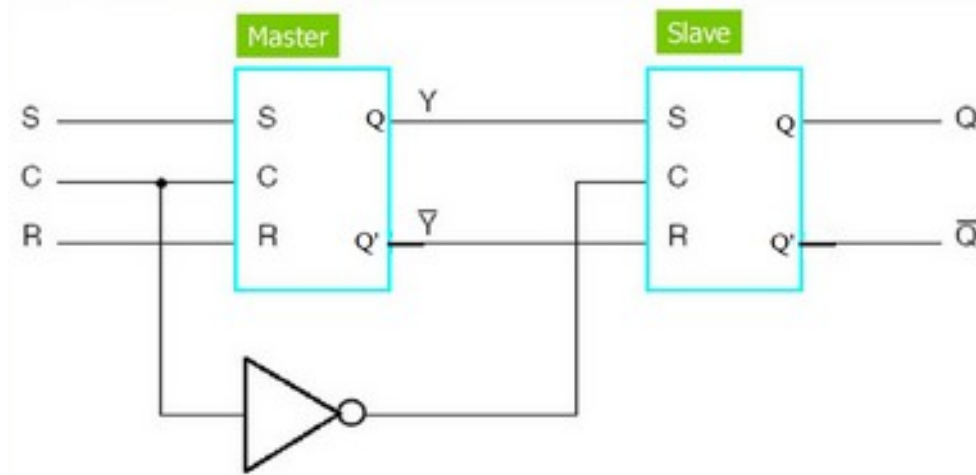


(a) D flip-flop with asynchronous clear

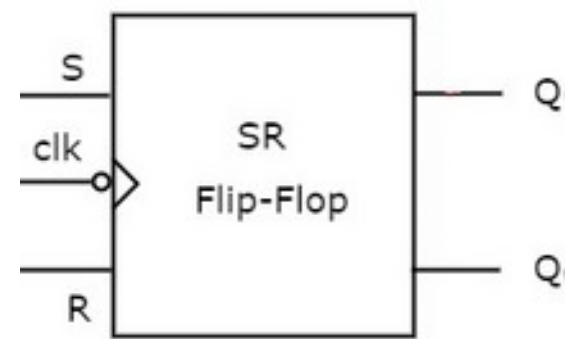
- If the Clear Signal is considered to be Asynchronous signal, then that means the Clear signal will not wait for clock (positive edge as per the diagram here), and Q will made zero (0) immediately without waiting for the clock's required edge. That is Clear signal is not synchronized with Clock signal.
- If the Clear Signal is considered to be Synchronous signal, then that means the Clear signal will wait clock till immediate positive edge (positive edge as per the diagram here) occurs on the clock and then Q will be made zero (0). That is Clear signal is synchronized with Clock signal.



Master Slave SR Flip Flop



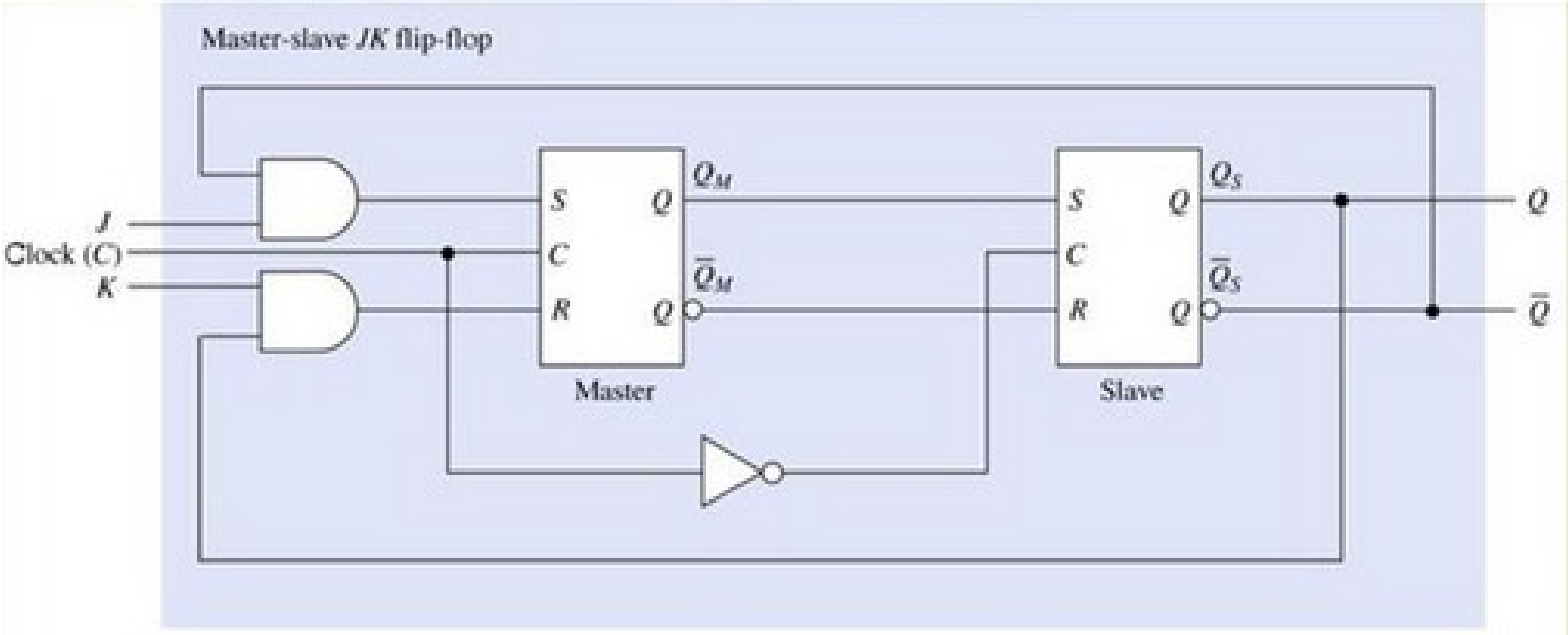
$C = \text{Clock} = \text{Clk}$



Clk	S	R	Q	Q'
0	x	x	Q	Q'
1	0	0	Q	Q
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

$C = \text{Clk} = \text{Clock}$	S	R	Q	Q'	Remarks
Negative edge on C	0	0	Q	Q'	Same as that of the previous value
Negative edge on C	0	1	0	1	Reset state
Negative edge on C	1	0	1	0	Set state
Negative edge on C	1	1	1	1	Invalid or Undefined
Any other value on C other than negative edge on the C	x	x	Q	Q'	Same as that of the previous value

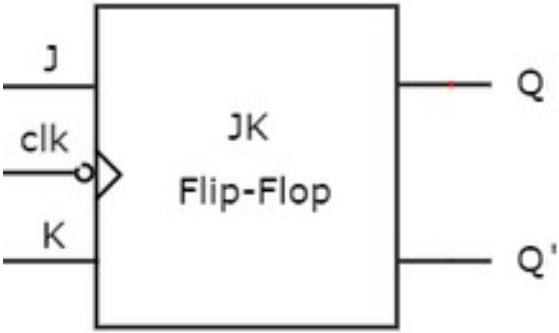
J K Flipflop (Master Slave)



Clock = C = Clk

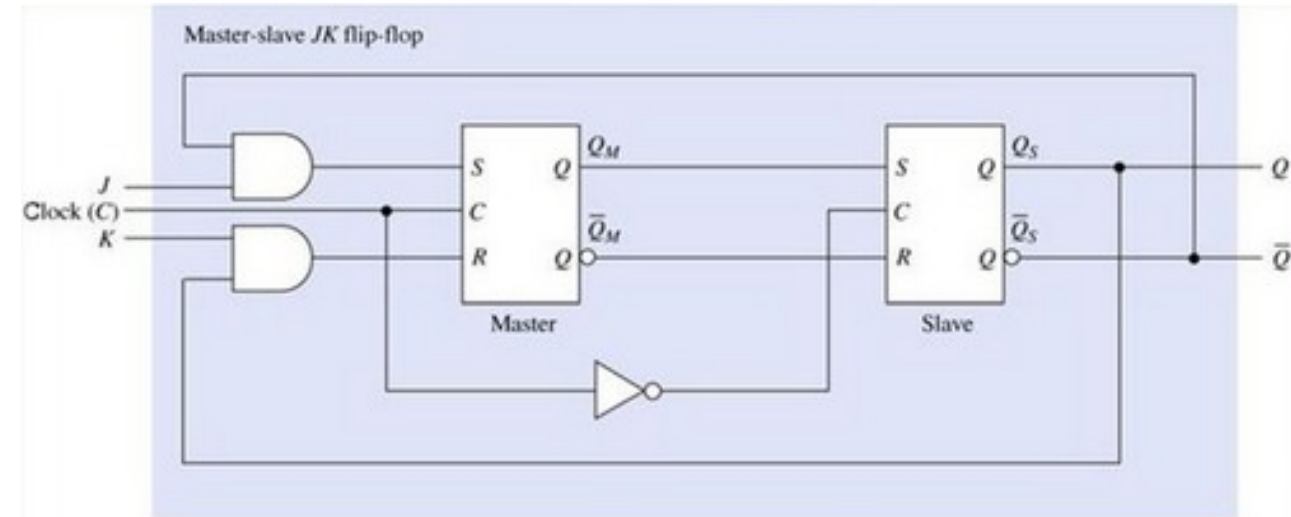
Clk	S	R	Q	Q'
0	x	x	Q	Q'
1	0	0	Q	Q
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

Truth Table of JK Flip Flop					
Clk	J	K	Q	Q'	Remarks
Negative edge on Clk	0	0	Q	Q'	Same as that of the previous value
Negative edge on Clk	0	1	0	1	
Negative edge on Clk	1	0	1	0	
Negative edge on Clk	1	1	Q'	Q	Invert of the previous value
Any other value on clock other than negative edge on clk	x	x	Q	Q'	Same as that of the previous value



J K Flipflop (Master Slave)

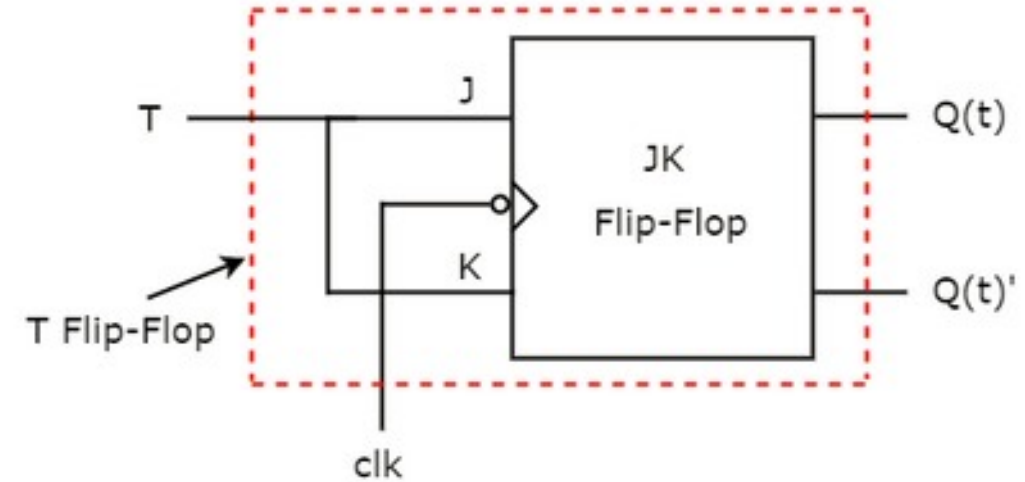
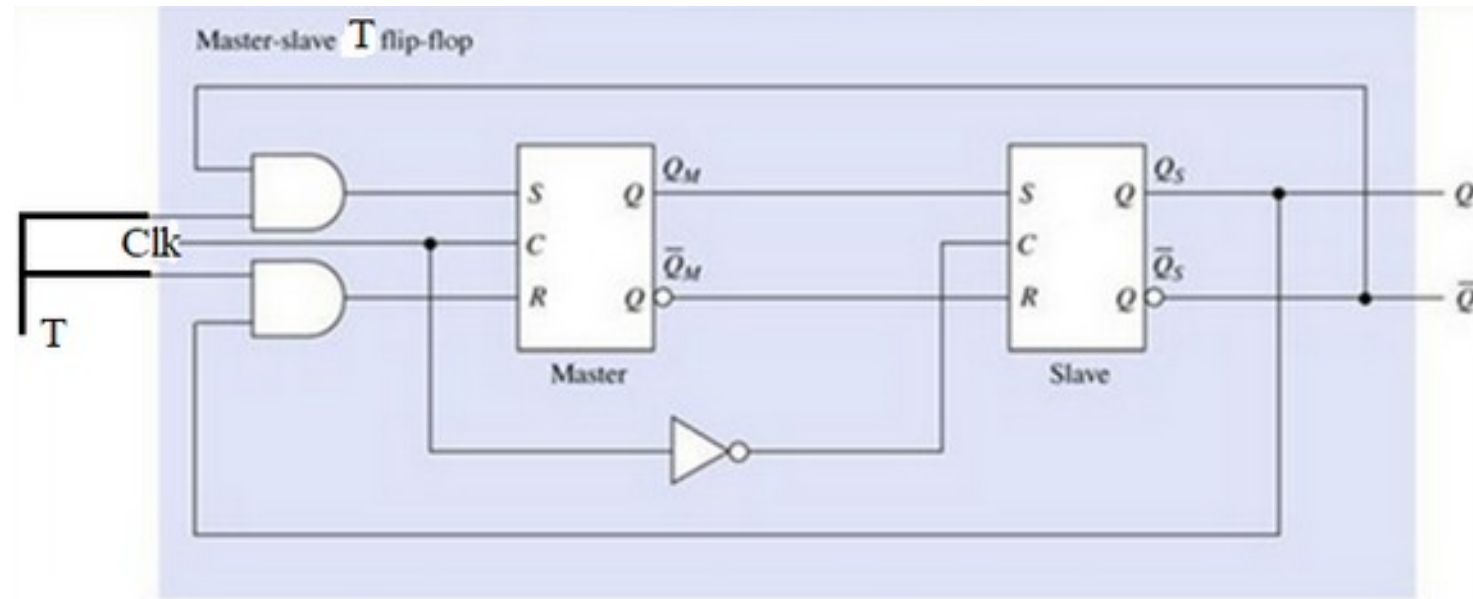
- When $\text{clk} = 1$ Master Latch is active and when $\text{clk} = 0$ Slave latch is active.
- a) **When $J = 0$ and $K = 0$** , Master Latch input $S = R = 0$ therefore Q_M and Q_M' will be same as that of the previous value. Therefore, based on previous value on $Q_M = 0/1$ and $Q_M' = 1/0$, the output $Q_S = 0/1$ and $Q_S' = 1/0$.
- b) **When $J = 0$ and $K = 1$** , Master Latch input $S = 0$ and R can not be defined hence we assume
 - Case i) If we assume at this time $Q = 0$ and $Q' = 1$, then $R = 0$ hence Q_M and Q_M' will have the same previous value $Q_M = 0$ and $Q_M' = 1$. This causes the Q_S and Q_S' to be 0 and 1 respectively (same as previous).
 - Case ii) If we assume at this time $Q = 1$ and $Q' = 0$, then $R = 1$ hence Q_M and Q_M' will have the value as $Q_M = 0$ and $Q_M' = 1$. This causes the Q_S and Q_S' to be 0 and 1 respectively.
 - Hence, we conclude that when $J = 0$ and $K = 1$, then $Q = 0$ and $Q' = 1$
- c) **When $J = 1$ and $K = 0$** , Master Latch input $R = 0$ and S can not be defined hence we assume
 - Case i) If we assume at this time $Q = 0$ and $Q' = 1$, then $S = 1$ hence $Q_M = 1$ and $Q_M' = 0$. This causes the Q_S and Q_S' to be 1 and 0 respectively.
 - Case ii) If we assume at this time $Q = 1$ and $Q' = 0$, then $S = 0$ hence Q_M and Q_M' will have the same previous value $Q_M = 1$ and $Q_M' = 0$. This causes the Q_S and Q_S' to be 1 and 0 respectively (same as previous).
 - Hence, we conclude that when $J = 1$ and $K = 0$, then $Q = 1$ and $Q' = 0$
- d) **When $J = 1$ and $K = 1$** , Master Latch input S and R inputs can not be defined hence we assume
 - Case i) If we assume at this time $Q = 0$ and $Q' = 1$, then $S = 1$ and $R = 0$, hence $Q_M = 1$ and $Q_M' = 0$. This causes the Q_S and Q_S' to be 1 and 0 respectively.
 - Case ii) If we assume at this time $Q = 1$ and $Q' = 0$, then $S = 0$ and $R = 1$, hence $Q_M = 0$ and $Q_M' = 1$. This causes the Q_S and Q_S' to be 0 and 1 respectively.
 - Hence, we conclude that when $J = 1$ and $K = 1$, then Q and Q' toggles with respect to the previous value and hence it is a valid output and removes the drawback of the Master Slave SR flipflop where for $S=R=1$, the output is invalid as discussed before.



Clk	S	R	Q	Q'
0	x	x	Q	Q'
1	0	0	Q	Q
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

Truth Table of JK Flip Flop				
Clk	J	K	Q	Q'
Negative edge on Clk	0	0	Q	Q'
Negative edge on Clk	0	1	0	1
Negative edge on Clk	1	0	1	0
Negative edge on Clk	1	1	Q'	Q
Any other value on clock other than negative edge on clk	x	x	Q	Q'

T (Toggle) Flip Flop



C = Clk = Clock	T	Q	Q'	Remarks
Negative edge on C	0	Q	Q'	Same as that of the previous value
Negative edge on C	1	Q'	Q	Toggles i. e. Invert of the previous value
Any other value on C other than negative edge on the C	x	Q	Q'	Same as that of the previous value

- Summary of Truth Table of SR, JK, D and T flipflops
 - Characteristic Table
 - Characteristic Equation
 - Excitation Table
- Characteristic Table = State Table
Function Table = Truth Table

Master-Slave JK Flip-Flop

Q_n	J_n	K_n	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

State Table



J_n	K_n	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

Characteristic Table

		$J_n K_n$			
Q_n		00	01	11	10
		0	0	1	1
	1	1	0	0	1

K-Map

$$Q_{n+1} = J_n \cdot \overline{Q_n} + \overline{K_n} \cdot Q_n$$

Characteristic Equation

Q_n	Q_{n+1}	J_n	K_n
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

Excitation Table

The d indicates a "don't care" term. That is, the signal value can be either a 0 or a 1 as it doesn't affect the next state.

Q_n	D_n	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

State Table

D_n	Q_{n+1}
0	0
1	1

Characteristic Table

	D_n	0	1
Q_n	0	0	1
	1	0	1

K-Map

$$Q_{n+1} = D_n$$

Characteristic Equation

Q_n	Q_{n+1}	D_n
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table

- As an exercise, derive the state table, K-map, and excitation table for an SR flip-flop from the characteristic table.

SR		S	R	Q	Q_{next}
		0	0	0	0
		0	0	1	1
		0	1	0	0
		0	1	1	0
		1	0	0	1
		1	0	1	1
		1	1	0	x
		1	1	1	x

	SR = 00	SR = 01	SR = 11	SR = 10
Q = 0	0	0	x	1
Q = 1	1	0	x	1

Characteristic Equation for SR ff = $Q_{next} = Q^+ = S + R'Q$

JK		J	K	Q	Q_{next}
		0	0	0	0
		0	0	1	1
		0	1	0	0
		0	1	1	0
		1	0	0	1
		1	0	1	1
		1	1	0	1
		1	1	1	0

	JK = 00	JK = 01	JK = 11	JK = 10
Q = 0	0	0	1	1
Q = 1	1	0	0	1

Characteristic Equation for JK ff = $Q_{next} = Q^+ = JQ' + K'Q$

D		D	Q	Q_{next}
		0	x	0
		1	x	1

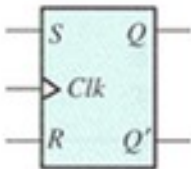
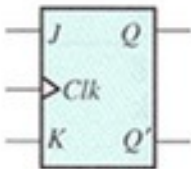
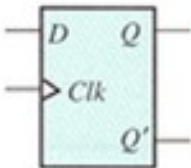
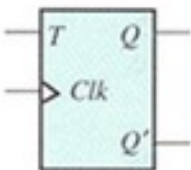
	D = 0	D = 1
Q = 0	0	1
Q = 1	0	1

Characteristic Equation for D ff = $Q_{next} = Q^+ = D$

T		T	Q	Q_{next}
		0	0	0
		0	1	1
		1	0	1
		1	1	0

	T = 0	T = 1
Q = 0	0	1
Q = 1	1	0

Characteristic Equation for T ff = $Q_{next} = Q^+ = T'Q + TQ' = T \oplus Q$

FLIP-FLOP NAME	FLIP-FLOP SYMBOL	CHARACTERISTIC TABLE	CHARACTERISTIC EQUATION	EXCITATION TABLE																																			
SR		<table><tr><th>S</th><th>R</th><th>Q(next)</th></tr><tr><td>0</td><td>0</td><td>Q</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>NA</td></tr></table>	S	R	Q(next)	0	0	Q	0	1	0	1	0	1	1	1	NA	$Q(next) = S + R'Q$	<table><tr><th>Q</th><th>Q(next)</th><th>S</th><th>R</th></tr><tr><td>0</td><td>0</td><td>0</td><td>X</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>X</td><td>0</td></tr></table>	Q	Q(next)	S	R	0	0	0	X	0	1	1	0	1	0	0	1	1	1	X	0
S	R	Q(next)																																					
0	0	Q																																					
0	1	0																																					
1	0	1																																					
1	1	NA																																					
Q	Q(next)	S	R																																				
0	0	0	X																																				
0	1	1	0																																				
1	0	0	1																																				
1	1	X	0																																				
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D		<table><tr><th>D</th><th>Q(next)</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr></table>	D	Q(next)	0	0	1	1	$Q(next) = D$	<table><tr><th>Q</th><th>Q(next)</th><th>D</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	Q	Q(next)	D	0	0	0	0	1	1	1	0	0	1	1	1														
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T		<table><tr><th>T</th><th>Q(next)</th></tr><tr><td>0</td><td>Q</td></tr><tr><td>1</td><td>Q'</td></tr></table>	T	Q(next)	0	Q	1	Q'	$Q(next) = TQ' + T'Q$	<table><tr><th>Q</th><th>Q(next)</th><th>T</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	Q	Q(next)	T	0	0	0	0	1	1	1	0	1	1	1	0														
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- Characteristic Table = State Table
Function Table = Truth Table